

Operations Research

Problem Set 4 – Duality of Linear Programs

Companion to Lecture 4. Quick checks, six problems, and full solutions from page 4.

Conventions as in Lecture 4. Canonical pair: $(P) \max c^\top x, Ax \leq b, x \geq 0$ and $(D) \min b^\top y, A^\top y \geq c, y \geq 0$. For a minimisation primal $\min c^\top x, Ax \geq b, x \geq 0$, the dual is $\max b^\top y, A^\top y \leq c, y \geq 0$ (derived in Set 2, Problem 3). Slacks: $s = b - Ax$ (primal), $r = A^\top y - c$ (dual); complementary slackness: $y_i s_i = 0$ and $x_j r_j = 0$.

Part A – Quick checks

Decide whether each statement is true or false, with a one-line justification. (Answers in the solutions.)

- A1. If x is feasible for (P) and y is feasible for (D) , then $c^\top x \leq b^\top y$.
- A2. If (P) is infeasible, then (D) is unbounded below.
- A3. A free primal variable produces a dual *equality* constraint.
- A4. If y^* is an optimal dual solution with $y_i^* > 0$, then the i -th primal constraint is tight at *every* optimal primal solution.
- A5. If the feasible region of (P) is unbounded, then (P) is unbounded above.
- A6. A primal with 3 constraints and 5 variables has a dual with 5 constraints and 3 variables.
- A7. If some feasible x, y satisfy $c^\top x = b^\top y$, then *all* optimal solutions of (P) have value $b^\top y$.
- A8. If (P) has an optimal solution, then (D) is feasible.

Part B – Problems

Problem 1 – Dualising with mixed signs

Consider

$$\begin{array}{llllll}
 \max & 2x_1 & - & x_2 & + & 3x_3 \\
 \text{s.t.} & x_1 & + & x_2 & + & x_3 & \leq & 6, \\
 & x_1 & & & - & x_3 & \geq & -2, \\
 & 2x_1 & + & x_2 & & & = & 4, \\
 & x_1 \geq 0, & x_2 \leq 0, & x_3 \in \mathbb{R}.
 \end{array}$$

- (a) Write the dual. For each dual variable give its sign and the primal row it comes from; for each dual constraint give the primal variable it comes from and explain its direction.
- (b) Show that the dual is infeasible.
- (c) What does (b) imply for the primal? Confirm it directly by exhibiting a feasible ray along which the objective tends to $+\infty$.

Problem 2 – How good is this plan?

A workshop chooses activity levels $x_1, x_2, x_3 \geq 0$:

$$\begin{array}{llllll} \max & 3x_1 & + & 2x_2 & + & 4x_3 \\ \text{s.t.} & x_1 & + & x_2 & + & 2x_3 & \leq & 4 & (R_1), \\ & 2x_1 & & & & + & 3x_3 & \leq & 5 & (R_2), \\ & & & x_2 & + & x_3 & \leq & 2 & (R_3). \end{array}$$

- Verify that $\hat{x} = (1, 1, 1)$ is feasible and compute its profit.
- By adding the three constraints with suitable nonnegative weights, prove by hand that no feasible plan earns more than 11. What does this say about $\hat{x}$?
- Write the dual (D).
- Use complementary slackness to decide whether $\hat{x}$ is optimal.
- Find optimal solutions x^* and y^* , verify them, and give the exact profit lost by $\hat{x}$.
- Is x^* the unique optimal plan? Which resource has zero price, and why?

Problem 3 – Complementary slackness as an optimality test

$$\begin{array}{llllll} \max & 5x_1 & + & 4x_2 & + & 3x_3 \\ \text{s.t.} & 2x_1 & + & 3x_2 & + & x_3 & \leq & 5, \\ & 4x_1 & + & x_2 & + & 2x_3 & \leq & 11, \\ & 3x_1 & + & 4x_2 & + & 2x_3 & \leq & 8, \\ & x_1, x_2, x_3 & \geq & 0. \end{array}$$

- Write the dual.
- A colleague claims $\hat{x} = (0, 1, 2)$ is optimal. Check feasibility, then use complementary slackness to test the claim.
- Test $\tilde{x} = (2, 0, 1)$ the same way. Conclude.
- Prove that the optimal plan is unique.
- Interpret the optimal dual values as prices: what is one extra unit of each resource worth?

Problem 4 – The four outcomes

For each LP, determine whether the primal is finite, unbounded or infeasible, do the same for its dual, and give a certificate for each claim (an optimal pair, an improving ray, or an infeasibility argument).

- $\max x_1 + x_2$ s.t. $x_1 - x_2 \leq 2, x \geq 0$.
- $\max x_1$ s.t. $x_1 + x_2 \leq 1, -x_1 - x_2 \leq -2, x \geq 0$.
- $\max 2x_1 + x_2$ s.t. $x_1 + x_2 \leq 3, x_1 \leq 2, x \geq 0$.
- Give an example (with proof) where both primal and dual are infeasible.

Summarise your findings in the primal/dual outcome table of the lecture.

Problem 5 – Short proofs

For the canonical pair (P) , (D) , prove:

- (a) If feasible x , y satisfy complementary slackness ($y_i s_i = 0$ for all i and $x_j r_j = 0$ for all j), then both are optimal.
- (b) The set of optimal solutions of (P) is convex.
- (c) If (P) has an optimal solution, then (D) is feasible.
- (d) If b is replaced by $2b$ and (P) remains feasible, then the optimal value of (P) doubles and the set of optimal dual solutions is unchanged.

Problem 6 – Prices in a minimisation

A feed must supply at least 4, 6 and 5 units of three nutrients. Two raw materials cost 3 and 2 MAD per kg and contain, per kg, $(1, 1, 2)$ and $(1, 3, 1)$ units of the nutrients.

- (a) Formulate the cost-minimising LP and write its dual. Interpret the dual variables.
- (b) Solve the primal graphically and find the optimal dual by complementary slackness. Verify optimality.
- (c) The requirement for nutrient 1 rises from 4 to 4.5. Using the dual prices, estimate the new minimum cost; then compute it exactly and compare.
- (d) A supplier offers a pre-mixed product containing $(1, 1, 1)$ units per kg at price p MAD. Using the dual solution, for which values of p is it worth buying? Confirm with an explicit plan for $p = 1.5$.

Solutions

Part A

- A1. **True.** Weak duality: $c^\top x \leq y^\top Ax \leq b^\top y$.
- A2. **False.** Both may be infeasible, e.g. the lecture's example $x_1 - x_2 \leq -1$, $-x_1 + x_2 \leq -1$ (see Problem 4(d)).
- A3. **True.** The combined inequality $\sum_j (A^\top y)_j x_j \leq b^\top y$ must dominate $c_j x_j$ for x_j of either sign, which forces $(A^\top y)_j = c_j$.
- A4. **True.** Any optimal primal x and any optimal dual y^* form an optimal pair, so complementary slackness $y_i^* s_i = 0$ holds; $y_i^* > 0$ forces $s_i = 0$.
- A5. **False.** $\max -x_1$ s.t. $x \geq 0$ has an unbounded feasible region and optimum 0. Unbounded region does not imply unbounded objective.
- A6. **True.** One dual variable per primal constraint, one dual constraint per primal variable.
- A7. **True.** By weak duality $b^\top y$ is an upper bound on every feasible value and it is attained, so it is the optimal value; every optimal solution has that value.
- A8. **True.** An optimal solution means (P) is feasible and bounded; strong duality then gives an optimal, in particular feasible, dual solution.

Solution 1(a)

Rows give variables. Row 1 is $\leq$ in a maximisation, so $y_1 \geq 0$ (a nonnegative weight preserves $\leq$). Row 2 is $\geq$, so $y_2 \leq 0$ (a nonpositive weight reverses $\geq$ into $\leq$). Row 3 is an equality, so $y_3 \in \mathbb{R}$ (any weight preserves an equality).

Columns give constraints. Let $d_j = \sum_i a_{ij} y_i$ be the coefficient of x_j in the weighted sum. We need $c_j x_j \leq d_j x_j$ for every admissible x_j .

- $x_1 \geq 0$ (column $(1, 1, 2)$): need $d_1 \geq c_1$, i.e. $y_1 + y_2 + 2y_3 \geq 2$.
- $x_2 \leq 0$ (column $(1, 0, 1)$): multiplying by a nonpositive number reverses the order, so need $d_2 \leq c_2$, i.e. $y_1 + y_3 \leq -1$.
- $x_3 \in \mathbb{R}$ (column $(1, -1, 0)$): need $d_3 = c_3$, i.e. $y_1 - y_2 = 3$.

Objective. The weighted right-hand side $6y_1 - 2y_2 + 4y_3$ is an upper bound on the profit; minimise it. The dual is

$$\begin{array}{llllll} \min & 6y_1 & - & 2y_2 & + & 4y_3 \\ \text{s.t.} & y_1 & + & y_2 & + & 2y_3 & \geq & 2 & (x_1 \geq 0), \\ & y_1 & & & + & y_3 & \leq & -1 & (x_2 \leq 0), \\ & y_1 & - & y_2 & & & = & 3 & (x_3 \text{ free}), \\ & y_1 \geq 0, & y_2 \leq 0, & y_3 \in \mathbb{R}. \end{array}$$

Solution 1(b)

Suppose y is dual feasible. From the equality, $y_2 = y_1 - 3$, and $y_2 \leq 0$ gives $y_1 \leq 3$. Substituting into the first constraint,

$$y_1 + (y_1 - 3) + 2y_3 \geq 2 \iff y_3 \geq \frac{5}{2} - y_1.$$

The second constraint says $y_3 \leq -1 - y_1$. Both together require $\frac{5}{2} - y_1 \leq -1 - y_1$, i.e. $\frac{5}{2} \leq -1$,

which is false. No dual feasible point exists.

Solution 1(c)

By the lecture's outcome table, an infeasible dual leaves two possibilities for the primal: unbounded or infeasible. Weak duality alone cannot decide which, so we look at the primal directly.

The primal is feasible: $x = (2, 0, 0)$ satisfies $2 \leq 6$, $2 \geq -2$, $4 = 4$ and the sign conditions. We exhibit a ray. For $t \geq 0$ set

$$x(t) = (2 + \frac{t}{2}, -t, 4 + \frac{t}{2}).$$

Signs: $x_1 \geq 0$, $x_2 \leq 0$, x_3 free. Constraints: row 1 gives $(2 + \frac{t}{2}) - t + (4 + \frac{t}{2}) = 6 \leq 6$; row 2 gives $(2 + \frac{t}{2}) - (4 + \frac{t}{2}) = -2 \geq -2$; row 3 gives $2(2 + \frac{t}{2}) - t = 4$. So $x(t)$ is feasible for every $t \geq 0$, and

$$2x_1 - x_2 + 3x_3 = (4 + t) + t + (12 + \frac{3t}{2}) = 16 + \frac{7}{2}t \rightarrow +\infty.$$

The primal is unbounded above, consistent with Problem 3(c) of Set 3: an unbounded primal must have an infeasible dual.

Solution 2(a)

$R_1: 1 + 1 + 2 = 4 \leq 4$; $R_2: 2 + 3 = 5 \leq 5$; $R_3: 1 + 1 = 2 \leq 2$. All three are tight, and $\hat{x} \geq 0$. Profit $3 + 2 + 4 = 9$.

Solution 2(b)

Add the three constraints with weight 1 each:

$$(x_1 + x_2 + 2x_3) + (2x_1 + 3x_3) + (x_2 + x_3) \leq 4 + 5 + 2 \iff 3x_1 + 2x_2 + 6x_3 \leq 11.$$

Since $x_3 \geq 0$, $3x_1 + 2x_2 + 4x_3 \leq 3x_1 + 2x_2 + 6x_3 \leq 11$ for every feasible x . Hence $9 \leq p^* \leq 11$: the plan $\hat{x}$ is at most 2 below the optimum. (The weights $(1, 1, 1)$ are a dual feasible point, see (c).)

Solution 2(c)

One variable per resource, one constraint per activity (columns $(1, 2, 0)$, $(1, 0, 1)$, $(2, 3, 1)$):

$$\begin{array}{llllll} \min & 4y_1 & + & 5y_2 & + & 2y_3 \\ \text{s.t.} & y_1 & + & 2y_2 & & & \geq 3 & (x_1), \\ & y_1 & & & + & y_3 & \geq 2 & (x_2), \\ & 2y_1 & + & 3y_2 & + & y_3 & \geq 4 & (x_3), \\ & y_1, y_2, y_3 & \geq 0. \end{array}$$

Solution 2(d)

If $\hat{x}$ were optimal, some optimal y would satisfy complementary slackness with it. All three coordinates of $\hat{x}$ are positive, so all three dual constraints must be tight:

$$y_1 + 2y_2 = 3, \quad y_1 + y_3 = 2, \quad 2y_1 + 3y_2 + y_3 = 4.$$

From the first two, $y_2 = \frac{3-y_1}{2}$ and $y_3 = 2 - y_1$. Substituting into the third: $2y_1 + \frac{9-3y_1}{2} + 2 - y_1 = 4$, i.e. $-\frac{1}{2}y_1 + \frac{13}{2} = 4$, so $y_1 = 5$, $y_2 = -1$, $y_3 = -3$. This y violates $y \geq 0$, so no dual feasible solution satisfies complementary slackness with $\hat{x}$. Therefore $\hat{x}$ is **not optimal**. (The primal

slacks were all zero, so they imposed no condition; the test was decided entirely by the dual side.)

Solution 2(e)

Guess from the economics. Activity 3 uses a lot of every resource for a profit of 4; try dropping it. With $x_3 = 0$ the problem is $\max 3x_1 + 2x_2$ s.t. $x_1 + x_2 \leq 4$, $2x_1 \leq 5$, $x_2 \leq 2$, whose best point is $x_1 = \frac{5}{2}$, $x_2 = \frac{3}{2}$ (both R_1 and R_2 tight). Candidate $x^* = (\frac{5}{2}, \frac{3}{2}, 0)$, profit $\frac{15}{2} + 3 = \frac{21}{2}$. Feasibility: R_1 : $4 \leq 4$; R_2 : $5 \leq 5$; R_3 : $\frac{3}{2} \leq 2$.

Dual by complementary slackness. R_3 has slack $\frac{1}{2} > 0$, so $y_3 = 0$. $x_1, x_2 > 0$ force their dual constraints tight: $y_1 + 2y_2 = 3$ and $y_1 + y_3 = 2$, hence $y_1 = 2$, $y_2 = \frac{1}{2}$. Candidate $y^* = (2, \frac{1}{2}, 0)$.

Verification. $y^* \geq 0$; dual constraints: $2 + 1 = 3 \geq 3$, $2 + 0 = 2 \geq 2$, $4 + \frac{3}{2} + 0 = \frac{11}{2} \geq 4$. Dual value $8 + \frac{5}{2} + 0 = \frac{21}{2} = c^T x^*$. Feasible pair with equal values: both optimal. In words, y^* says $2R_1 + \frac{1}{2}R_2$ reads $3x_1 + 2x_2 + \frac{11}{2}x_3 \leq \frac{21}{2}$, which dominates the profit. The exact loss of $\hat{x}$ is $\frac{21}{2} - 9 = \frac{3}{2}$, inside the interval $[0, 2]$ of (b).

Solution 2(f)

Let x be any optimal plan; it forms an optimal pair with y^* , so complementary slackness holds. $y_1^* = 2 > 0$ and $y_2^* = \frac{1}{2} > 0$ force R_1 and R_2 tight. The dual slack of x_3 is $r_3 = \frac{11}{2} - 4 = \frac{3}{2} > 0$, which forces $x_3 = 0$. Then $x_1 + x_2 = 4$ and $2x_1 = 5$ give $x = (\frac{5}{2}, \frac{3}{2}, 0)$. The optimum is unique. Resource 3 has price $y_3^* = 0$ because it is not fully used at the optimum ($s_3 = \frac{1}{2}$): an extra unit of it would change nothing. Resource 1 is worth 2 per unit and resource 2 is worth $\frac{1}{2}$ per unit, at the margin.

Solution 3(a)

Columns $(2, 4, 3)$, $(3, 1, 4)$, $(1, 2, 2)$:

$$\min 5y_1 + 11y_2 + 8y_3 \quad \text{s.t.} \quad 2y_1 + 4y_2 + 3y_3 \geq 5, \quad 3y_1 + y_2 + 4y_3 \geq 4, \quad y_1 + 2y_2 + 2y_3 \geq 3, \quad y \geq 0.$$

Solution 3(b)

Feasibility of $\hat{x} = (0, 1, 2)$: row 1: $3 + 2 = 5 \leq 5$ (tight); row 2: $1 + 4 = 5 \leq 11$ (slack 6); row 3: $4 + 4 = 8 \leq 8$ (tight). Value $4 + 6 = 10$.

Complementary slackness: slack $s_2 = 6 > 0$ forces $y_2 = 0$; $x_2 > 0$ and $x_3 > 0$ force the second and third dual constraints tight:

$$3y_1 + 4y_3 = 4, \quad y_1 + 2y_3 = 3.$$

From the second, $y_1 = 3 - 2y_3$; then $9 - 6y_3 + 4y_3 = 4$ gives $y_3 = \frac{5}{2}$ and $y_1 = -2 < 0$. No nonnegative y satisfies the conditions, so $\hat{x}$ is **not optimal**.

Solution 3(c)

Feasibility of $\tilde{x} = (2, 0, 1)$: row 1: $4 + 1 = 5$ (tight); row 2: $8 + 2 = 10 \leq 11$ (slack 1); row 3: $6 + 2 = 8$ (tight). Value $10 + 3 = 13$.

Complementary slackness: $s_2 = 1 > 0$ forces $y_2 = 0$; $x_1, x_3 > 0$ force

$$2y_1 + 3y_3 = 5, \quad y_1 + 2y_3 = 3 \implies y_1 = 3 - 2y_3, \quad 6 - 4y_3 + 3y_3 = 5, \quad y_3 = 1, \quad y_1 = 1.$$

Candidate $y = (1, 0, 1)$. Check the remaining dual constraint: $3 + 0 + 4 = 7 \geq 4$. So y is dual feasible, with dual value $5 + 0 + 8 = 13 = c^T \tilde{x}$. **Conclusion:** $\tilde{x}$ and y are both optimal; the

optimal value is 13.

Solution 3(d)

Let x be any optimal plan. With the optimal dual $y = (1, 0, 1)$, complementary slackness gives: $y_1 > 0 \Rightarrow$ row 1 tight; $y_3 > 0 \Rightarrow$ row 3 tight; the dual slack of x_2 is $r_2 = 7 - 4 = 3 > 0 \Rightarrow x_2 = 0$. Then $2x_1 + x_3 = 5$ and $3x_1 + 2x_3 = 8$, whose unique solution is $x_1 = 2$, $x_3 = 1$. Hence $x = \tilde{x}$: the optimum is unique.

Solution 3(e)

$y_1 = 1$: one more unit of resource 1 raises the optimal value by 1 (for small changes that keep the same tight constraints). $y_3 = 1$: likewise for resource 3. $y_2 = 0$: resource 2 is not fully used (one unit is left over), so an extra unit is worth nothing. These are the *shadow prices*; the dual objective $5y_1 + 11y_2 + 8y_3 = 13$ is the total value of the resources at those prices, and strong duality says it equals the maximum profit.

Solution 4(a)

Primal unbounded. $x(t) = (t, t)$, $t \geq 0$, satisfies $t - t = 0 \leq 2$ and $x \geq 0$, with objective $2t \rightarrow +\infty$. *Dual infeasible.* The dual is $\min 2y$ s.t. $y \geq 1$, $-y \geq 1$, $y \geq 0$; the second constraint requires $y \leq -1$, contradicting $y \geq 0$.

Solution 4(b)

Primal infeasible. Adding the two constraints gives $0 \leq -1$. *Dual unbounded below.* The dual is $\min y_1 - 2y_2$ s.t. $y_1 - y_2 \geq 1$, $y_1 - y_2 \geq 0$, $y \geq 0$. For $t \geq 0$, $y(t) = (1 + t, t)$ is feasible and its objective is $1 + t - 2t = 1 - t \rightarrow -\infty$.

Solution 4(c)

Both finite, value 5. $x^* = (2, 1)$ is feasible ($3 \leq 3$, $2 \leq 2$) with value 5. The dual is $\min 3y_1 + 2y_2$ s.t. $y_1 + y_2 \geq 2$, $y_1 \geq 1$, $y \geq 0$; $y^* = (1, 1)$ is feasible with value 5. Equal values, so both optimal.

Solution 4(d)

Take $\max x_1 + x_2$ s.t. $x_1 - x_2 \leq -1$, $-x_1 + x_2 \leq -1$, $x \geq 0$. Adding the constraints gives $0 \leq -2$: primal infeasible. The dual is $\min -y_1 - y_2$ s.t. $y_1 - y_2 \geq 1$, $-y_1 + y_2 \geq 1$, $y \geq 0$; adding its constraints gives $0 \geq 2$: dual infeasible.

Problem	Primal	Dual
4(a)	unbounded	infeasible
4(b)	infeasible	unbounded
4(c)	finite, 5	finite, 5
4(d)	infeasible	infeasible

The only combination that never occurs is finite on one side with anything but finite on the other.

Solution 5(a)

For feasible x, y the gap identity of the lecture holds:

$$b^\top y - c^\top x = y^\top (b - Ax) + x^\top (A^\top y - c) = \sum_{i=1}^m y_i s_i + \sum_{j=1}^n x_j r_j.$$

(Expand: $y^\top b - y^\top Ax + x^\top A^\top y - x^\top c = b^\top y - c^\top x$, since $y^\top Ax = x^\top A^\top y$.) If every product is zero, the gap is zero, so $c^\top x = b^\top y$. By Problem 3(b) of Set 3 (feasible pair with equal values), both are optimal.

Solution 5(b)

Let x, x' be optimal with common value v^* , and $\lambda \in [0, 1]$, $z = \lambda x + (1 - \lambda)x'$. Then $z \geq 0$ and $Az = \lambda Ax + (1 - \lambda)Ax' \leq \lambda b + (1 - \lambda)b = b$, so z is feasible; and $c^\top z = \lambda v^* + (1 - \lambda)v^* = v^*$, so z is optimal. (Both uses of $\lambda, 1 - \lambda \geq 0$ are essential.)

Solution 5(c)

If x^* is optimal, then (P) is feasible and $c^\top x \leq c^\top x^*$ for all feasible x , so the objective is bounded above. By strong duality (D) has an optimal solution, which is in particular feasible.

Solution 5(d)

Dual. The dual feasible set $\{y : A^\top y \geq c, y \geq 0\}$ does not involve b , so it is unchanged. The new dual objective is $(2b)^\top y = 2b^\top y$, a positive multiple of the old one, so the two objectives have the same minimisers and the new minimum is twice the old. *Primal.* x is feasible for b iff $2x$ is feasible for $2b$ (since $Ax \leq b \iff A(2x) \leq 2b$ and $x \geq 0 \iff 2x \geq 0$), and $c^\top(2x) = 2c^\top x$. So the new maximum is $2 \max c^\top x$. (Strong duality gives the same conclusion from the dual side.)

Solution 6(a)

Let $x_1, x_2 \geq 0$ be kilograms of the two materials.

$$\begin{array}{ll} \min & 3x_1 + 2x_2 \\ \text{s.t.} & x_1 + x_2 \geq 4 \quad (N_1), \\ & x_1 + 3x_2 \geq 6 \quad (N_2), \\ & 2x_1 + x_2 \geq 5 \quad (N_3), \\ & x_1, x_2 \geq 0. \end{array} \qquad \begin{array}{ll} \max & 4y_1 + 6y_2 + 5y_3 \\ \text{s.t.} & y_1 + y_2 + 2y_3 \leq 3 \quad (x_1), \\ & y_1 + 3y_2 + y_3 \leq 2 \quad (x_2), \\ & y_1, y_2, y_3 \geq 0. \end{array}$$

The dual variable y_i is a price per unit of nutrient i . The dual constraints say that, at those prices, the nutrients contained in one kg of a material are worth no more than the material costs (otherwise the prices would be “too high” to be a valid lower bound). The dual maximises the value of the required nutrients: the largest amount a price-setter could charge for the requirement without any material being cheaper than its nutrient content.

Solution 6(b)

Primal. The feasible region is the intersection of three half-planes above the lines. Its vertices are the intersections that satisfy the remaining constraint: $N_1 \cap N_3$ at $(1, 3)$ (check N_2 : $1 + 9 = 10 \geq 6$); $N_1 \cap N_2$ at $(3, 1)$ (check N_3 : $7 \geq 5$); N_3 with the x_2 -axis at $(0, 5)$; N_2 with the x_1 -axis

at $(6, 0)$. ($N_2 \cap N_3 = (\frac{9}{5}, \frac{7}{5})$ violates N_1 .) Costs: 9, 11, 10, 18. Optimum $x^* = (1, 3)$, cost 9 MAD.

Dual by complementary slackness. At x^* : N_1 tight, N_3 tight, N_2 has surplus $10 - 6 = 4 > 0$, so $y_2 = 0$. Both $x_1, x_2 > 0$, so both dual constraints tight: $y_1 + 2y_3 = 3$, $y_1 + y_3 = 2$, giving $y_3 = 1$, $y_1 = 1$. $y^* = (1, 0, 1)$.

Verification. $y^* \geq 0$, constraints $1 + 0 + 2 = 3 \leq 3$ and $1 + 0 + 1 = 2 \leq 2$. Dual value $4 + 0 + 5 = 9 =$ primal cost. Both optimal.

Solution 6(c)

Estimate. The price of nutrient 1 is $y_1^* = 1$ MAD per unit, so raising the requirement by 0.5 should add about $0.5 \times 1 = 0.5$: new cost ≈ 9.5 .

Exact. With N_1 : $x_1 + x_2 \geq 4.5$, the same two constraints are tight at the new optimum: $x_1 + x_2 = 4.5$, $2x_1 + x_2 = 5$ give $x = (\frac{1}{2}, 4)$. Check N_2 : $\frac{1}{2} + 12 \geq 6$. Cost $\frac{3}{2} + 8 = 9.5$. The estimate is exact, because the optimal basis (the set of tight constraints) did not change; the dual solution is still $(1, 0, 1)$ and still feasible.

Solution 6(d)

At the optimal prices the nutrient content of one kg of the pre-mix is worth $y^{*\top}(1, 1, 1) = 1 + 0 + 1 = 2$ MAD. If $p \geq 2$ the pre-mix costs at least what its nutrients are worth and cannot lower the cost: with a third variable x_3 , y^* remains dual feasible (its new constraint reads $y_1 + y_2 + y_3 \leq p$, i.e. $2 \leq p$) with value 9, so 9 is still a lower bound and x^* stays optimal. If $p < 2$, y^* violates the new dual constraint, so the old certificate is gone and the optimum can drop.

Confirmation for $p = 1.5$. The plan $(x_1, x_2, x_3) = (0, 2, 3)$ supplies $(2 + 3, 6 + 3, 2 + 3) = (5, 9, 5) \geq (4, 6, 5)$ and costs $0 + 4 + 4.5 = 8.5 < 9$. Buying the pre-mix at 1.5 MAD is worth it.

Remark

The recurring method of this set: feasibility of a candidate is checked on the primal side, then complementary slackness turns the candidate into a small linear system for y . If that system has a nonnegative, dual-feasible solution with the same objective value, the candidate is optimal and y is its certificate; if it does not, the candidate is not optimal. Uniqueness comes from reading the same conditions backwards with the optimal y fixed.